

Choosing a Method and Reading the Roots

Square roots, factoring, the quadratic formula, and what the roots mean in a story

- Three methods are in play, and **nobody tells you which to use**: take square roots when the squared part is already alone; factor when the expression factors quickly; use the quadratic formula when it does not.
 - Before each problem, write one short line saying which method you picked. That line is part of the practice.
 - “Solve” means **list every solution**, unless the problem says otherwise.
 - In the story problems, solve the equation first and interpret afterwards — a solution can be perfectly correct and still not answer the question that was asked.
-

1. Solve: $x^2 = 81$

Answer: _____

2. Solve: $x^2 - 7x + 10 = 0$

Answer: _____

3. Solve: $(x - 3)^2 = 16$

Answer: _____

4. Use the quadratic formula on $x^2 + 4x - 6 = 0$. Give the *larger* solution, rounded to two decimal places.

Answer: _____

5. Solve: $2x^2 - 8x = 0$

Answer: _____

6. A ball is kicked straight up from the ground. Its height above the ground, in feet, after t seconds is $h = -16t^2 + 32t$.

(a) Solve $-16t^2 + 32t = 0$ and list both solutions, in seconds.

(b) In the space above, say which solution tells you when the ball lands, and explain what the other solution describes.

Answer: _____

7. Solve: $3x^2 - 5x - 2 = 0$

Answer: _____

8. A rectangular vegetable garden is 4 metres longer than it is wide. Its area is 45 square metres. Writing w for the width in metres, the area gives $w(w + 4) = 45$.

(a) Solve $w(w + 4) = 45$ and list both solutions.

(b) In the space above, state the width of the garden in metres and explain why the other solution is rejected.

Answer: _____

9. Solve: $2(x + 1)^2 = 50$

Answer: _____

10. Use the quadratic formula on $5x^2 - 3x - 4 = 0$. Give the *smaller* solution, rounded to two decimal places.

Answer: _____

11. Solve: $x^3 - 9x = 0$

Answer: _____

12. Synthesis challenge. A drama club sells tickets to its play. When the ticket price is x dollars, the club's profit for the night, in dollars, is $P = -x^2 + 24x - 108$.

- (a) Solve $-x^2 + 24x - 108 = 0$ and list both solutions.
- (b) In the space above, explain what those two ticket prices mean for the club, and say which prices give the club a profit above zero.

Answer: _____